PART THREE

Quantum Optics

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LIGHT QUANTA AND THEIR ORIGIN

plasma state of our sun and the distant stars, at high temperatures, are certainly the most prominent light sources in the universe. The fact that so many of the brightest stars emit the same spectra that we observe in our laboratories is direct evidence that light throughout the universe comes from the same chemical elements we find on the earth.

29.1 THE BOHR ATOM

$$m\frac{v^2}{r} = k\frac{Ze^2}{r^2} \qquad (29a)$$

Centripetal force = electrostatic attraction

Bohr adopted this relation as his *first assumption* and then introduced the *quantum theory*. His *second assumption* says that the angular momentum of the electron mvr must always be equal to a whole number of units of $h/2\pi$

- $$mvr = n\hbar \qquad (29b)$$

where m is the electron mass, $\hbar$ is Planck's constant of action (h divided by 2π), first introduced in 1905 by Max Planck to derive the law of thermal radiation, and n is a whole number called the *principal quantum number*;

This means that the electron is not free to move in just any orbit, like a satellite in classical mechanics, but only in certain quantized orbits. By combining Eqs. (29a) and (29b) and solving for the orbit radius we obtain

$$r = n^2 \frac{\hbar^2}{me^2Zk} = n^2(0.529177 \times 10^{-10}) \quad \text{m} \qquad (29c)$$

and by solving for the orbital speed v we obtain

$$v = \frac{1}{n}\frac{e^2Zk}{\hbar} = \frac{1}{n}(2.18768 \times 10^6) \quad \text{m/s} \qquad (29d)$$

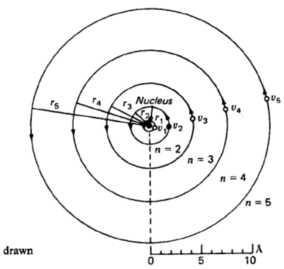

FIGURE 29B
Bohr circular orbits of hydrogen drawn to scale.

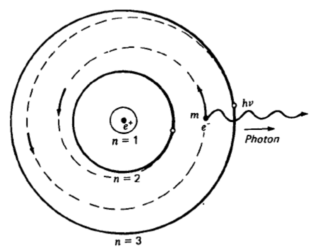

FIGURE 29C
Bohr's quantum theory of the radiation of light from a hydrogen atom.

- $$h\nu = E_i - E_f \qquad (29e)$$

where E_i is the *total energy in the initial orbit*, E_f is the *total energy in the final orbit*, h is again Planck's constant, and ν is the frequency of the emitted light.

By combining the three equations (29a), (29b), and (29e) and inserting the known values of the atomic constants, Bohr derived the following equation for all of the frequencies of light emitted by free hydrogen atoms:

$$\nu = 3.28984 \times 10^{15}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) \quad \text{Hz} \qquad (29f)$$

where n_i and n_f are the principal quantum numbers of the *initial* and *final orbits*. If we introduce the wave equation

$$c = \nu\lambda \qquad (29g)$$

and replace ν by c/λ, we obtain for the wavelengths of light*

- $$\lambda = 911.503 \frac{n_i^2 \times n_f^2}{n_i^2 - n_f^2} \quad \text{Å} \qquad (29h)$$

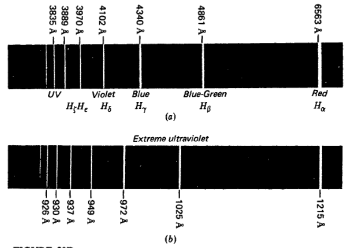

FIGURE 29D
The spectrum of the hydrogen atom: (a) the Balmer series and (b) the Lyman series.

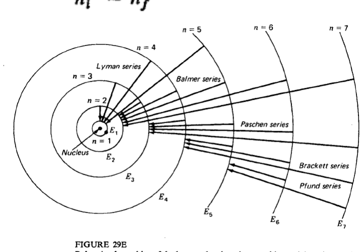

FIGURE 29E
Bohr circular orbits of hydrogen showing the transitions giving rise to the emitted light waves, or photons, of different frequency.

29.2 ENERGY LEVELS

The total energy E_{tot} of the electron in each of the Bohr orbits can be calculated from Bohr's first two postulates, Eqs. (29a) and (29b). Classically the potential energy E_{pot} is electrical in concept and is given by

$$E_{pot} = -k\frac{Ze^2}{r}$$

The kinetic energy, on the other hand, is mechanical, and is given by

$$E_{kin} = \tfrac{1}{2}mv^2 = k\frac{Ze^2}{2r}$$

Adding these two energies,

$$E_{tot} = -\frac{me^4Z^2k^2}{2n^2\hbar^2} \qquad (29i)$$

$$E_i - E_f = -R\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right) \qquad (29l) \qquad R = \frac{me^4Z^2k^2}{2\hbar^2} = 2.179350 \times 10^{-18}\ \mathrm{J}$$

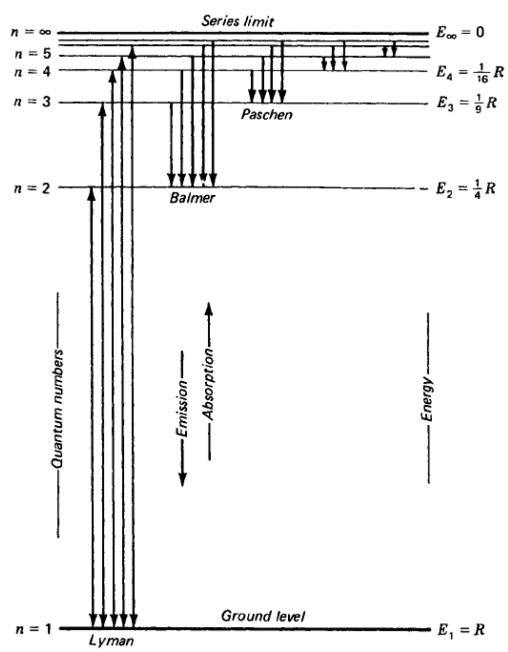

FIGURE 29F
Energy level diagram for the hydrogen atom. Vertical arrows represent electron transitions.

29.6 THE SPECTRUM OF SODIUM

Except for elements in the first two columns of the periodic table, the spectra of all elements are quite complex

The spectra of the alkali metals Li, Na, Mg, Ca, Sr, Ba, and Ra are relatively simple compared with those elements near the center of the periodic table.

The sodium energy level diagram in Fig. 29N shows the normal state, or ground state, as 3^2S and the succeeding excited states as 3^2P, 4^2S, 3^2D, 4^2P, etc. These level designations correspond to the orbit designations $3s$, $3p$, $4s$, $3d$, $4p$, etc. The superscript 2 indicates that all levels, with the exception of S states, are doublets. This doubling is due to the spinning of the electron and results in the doubling of all lines in all series.

Transitions from the two 3^2P levels to the ground state 3^2S give rise to the most prominent lines, the yellow D lines, account for the yellow color of all sodium lamps

resonance lines

At relatively low temperatures nearly all sodium atoms are in their ground state. As the temperature is raised, more and faster collisions occur between atoms and more have their valence electron bumped into excited states, with the subsequent emission of light.

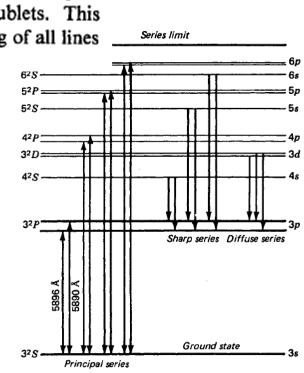

FIGURE 29N
An energy level diagram for the sodium atom, $Z = 11$, showing transitions for first members of the sharp, principal, and diffuse series.

29.8 METASTABLE STATES

the selection rules $\Delta n = 0, \pm 1, \pm 2, \pm 3, \pm 4, \cdots$ (29o)

$\Delta l = \pm 1$ only

for two electron systems

$\Delta l_1 = \pm 1$ and $\Delta l_2 = 0, \pm 2$ (29p)

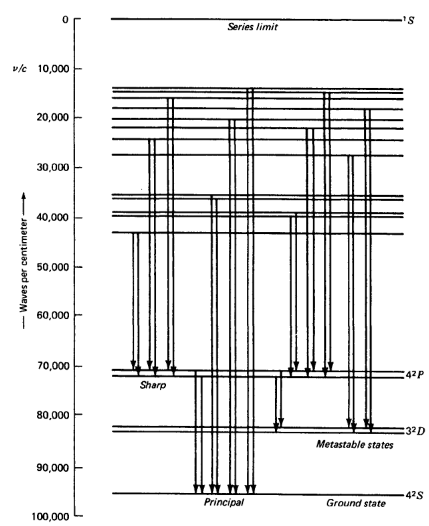

FIGURE 29Q
Energy level diagram of ionized calcium, showing the existence of metastable states.

29.9 OPTICAL PUMPING

Nearly all atoms in solids, liquids, or gases at near absolute zero are in their ground state. As the temperature is raised, by some form of energy input, more and more electrons are bumped into excited states. The populations of electrons in the higher energy levels increases at the expense of those that were in the ground level.

At any constant temperature a steady state will exist, and just as many electrons will be jumping into any level as will be jumping out of that level.

If a metastable state exists, the situation is different. As atoms are excited to higher levels, more and more of them get caught in the metastable level and relatively few of them get out except through mechanical collisions with other atoms.

The average populations of atoms in the metastable levels may be thousands and even millions of times that of any other, except the ground state. If they exceed the number in the ground state, it is called *population inversion.*

By shining light of a higher energy $h\nu$ than that required to excite an electron from the ground state to a metastable level, atoms can be *pumped* into that state by light absorption. The stronger the light source, the greater the number of electrons getting to the upper levels and then dropping back into the trap. This process is called *optical pumping.*

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LASERS

The name *laser* is an acronym of light amplification by stimulated emission of radiation. A laser is a device that produces an intense, concentrated, and highly parallel beam of coherent light. So parallel is the beam from a visible light laser 10 cm in diameter that at the moon's surface 384,000 km away the beam is no more than 5 km wide.

30.1 STIMULATED EMISSION

There are at least 10 basic principles involved in the operation of most lasers:

1 Metastable states
2 Optical pumping
3 Fluorescence
4 Population inversion
5 Resonance
6 Stimulated emission
7 Coherence
8 Polarization
9 Fabry-Perot interferometry
10 Cavity oscillation

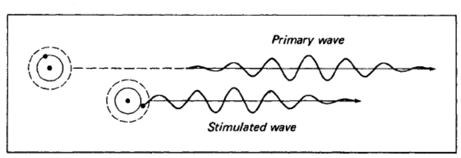

FIGURE 30A
The principle of stimulated emission of light from an atom. Both waves have the same wavelength λ and are in phase and vibrating in parallel planes.

30.2 LASER DESIGN

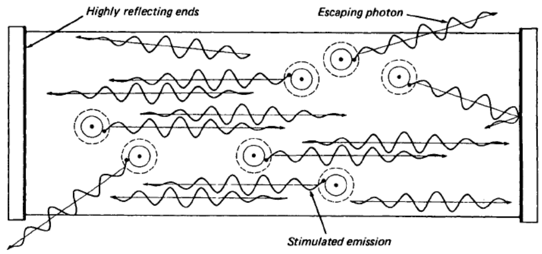

FIGURE 30B
A laser cavity with highly reflecting ends, showing the stimulated emission of light and the escape of some primary photons through the side walls.

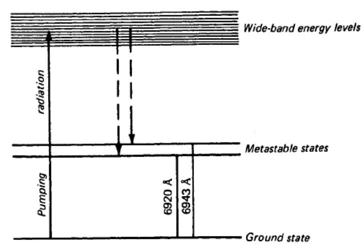

FIGURE 30C
Energy level diagram for a ruby crystal.

30.11 LASER SAFETY

Laser light varies in intensity from a fraction of a milliwatt for an inexpensive He–Ne laser to many kilowatts for a CO_2 laser. Laser injuries have been few, and their dangers highly debatable. However, the greatest danger is the inadvertent direction of an undiverged laser beam directly into the eye.

The weak beam from a $\frac{1}{2}$-mW continuous He–Ne laser is probably of little danger, since eyelids can close upon sudden exposure. More intense beams, and especially pulsed beams, can cause serious injury, due primarily to the ability of the eye to focus the parallel beam onto a small area of the retina.

Good safety practice in the presence of high-powered lasers involves the use of filtering glasses and shields and awareness that a laser beam incident upon a specular reflecting surface can redirect the beam undiminished in intensity.

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HOLOGRAPHY

The term *holography* comes from the Greek meaning *whole writing*. It is a two-step process by which (1) an object illuminated by coherent light is made to produce interference fringes in a photosensitive medium, such as a photographic emulsion, and (2) reillumination of the developed interference pattern by light of the same wavelength produces a three-dimensional image of the original object. The viewed images seen by this process have the appearance of the original object, including the differences in perspective one obtains with a change of the viewer's observing position—a full three-dimensional image.

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MAGNETO-OPTICS AND ELECTRO-OPTICS

of light, there is a group of optical experiments which demonstrates the interaction between light and matter when the latter is subjected to a strong external magnetic or electric field. In this group of experiments those which depend for their action on an applied magnetic field are classed under *magneto-optics* and those which depend for their action on an electric field are classed under *electro-optics.*

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THE DUAL NATURE OF LIGHT

33.2 EVIDENCE FOR LIGHT QUANTA

energy $h\nu$ are known as light quanta or *photons.*

the *photoelectric effect*

$E = h\nu - k$ *Energy of photoelectrons*   (33a)

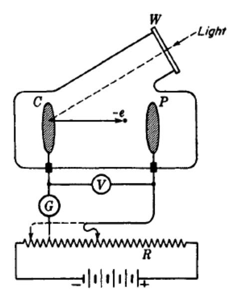

FIGURE 33A
Experimental arrangement for studying the photoelectric effect.

The *Compton effect* is observed in X rays that are scattered at an angle θ from a scatterer S consisting of some light element like carbon [see Fig. 33B(a)]. A narrow beam is defined by two lead slits and caused to fall on a crystal C. This diffracts the X rays to the photographic plate P, and by suitably turning the crystal about an axis perpendicular to the plane of the figure a spectrum can be photographed. For each monochromatic line present in the original X rays, the spectrum of the scattered rays shows a line shifted to longer wavelengths, the shift increasing with the scattering angle θ according to the equation

$$\Delta\lambda = \frac{c}{\nu'} - \frac{c}{\nu} = \frac{h}{m_0 c}(1 - \cos\theta) \qquad \textit{Compton shift} \qquad (33b)$$

where m_0 is the mass of an electron at rest and h/m_0c is called the *Compton wavelength.*

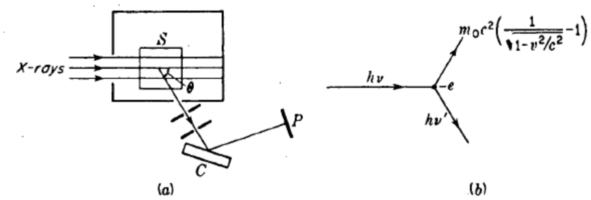

FIGURE 33B
The Compton effect: (a) method of observation; (b) energies of the incident photon, scattered photon, and recoil electron.

33.3 ENERGY, MOMENTUM, AND VELOCITY OF PHOTONS

$$E = h\nu \qquad \textit{Energy of a photon} \qquad (33c)$$

$$p = mc = \frac{h\nu}{c} = \frac{h}{\lambda} \qquad \textit{Momentum of a photon}^* \qquad (33e)$$

Velocity of a photon $= c$ (30f)

An experiment can measure both, but only within certain limits of accuracy. These limits are specified by the principle of indeterminacy (often called the *uncertainty principle*), according to which

- $$\Delta p_y \Delta y \gtrsim \frac{h}{2\pi} \qquad (33h)$$

* Einstein's general theory of relativity postulates an increase in momentum and mass of a photon as it passes through a strong gravitational field like that close to the sun.

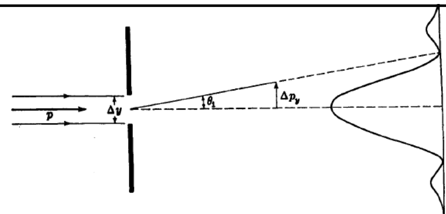

FIGURE 33D
The uncertainty principle applied to the momentum of a photon when it is diffracted by a single slit.

angle is given by

$$\sin \theta_1 = \frac{\lambda}{\Delta y} \qquad (33i)$$

The corresponding uncertainty in the momentum is

$$\Delta p_y = p \sin \theta_1 = \frac{p\lambda}{\Delta y} \qquad (33j)$$

Introducing the value for the momentum p given by the de Broglie relation, Eq. (33e), we find

$$\Delta p_y = \frac{h}{\lambda}\frac{\lambda}{\Delta y} = \frac{h}{\Delta y} \qquad (33k)$$

This gives $\Delta p_y \, \Delta y = h$, but it will be seen that since the probability of the photon striking the center of the pattern is greatest, the uncertainty in p_y is not so large as is indicated by Eq. (33k) and hence that our result is consistent with the principle of indeterminacy

$$\Delta p_y \, \Delta y \gtrsim \frac{h}{2\pi} \qquad (33h)$$

33.7 COMPLEMENTARITY

We owe to Bohr the interpretation of Heisenberg's principle in such a way as to make clear the fundamental limitations on the accuracy of measurement and their bearing on our views as to the nature of light and matter. According to the principle of complementarity stated by Bohr in 1928, the wave and corpuscular descriptions are merely complementary ways of regarding the same phenomenon. That is, to obtain the complete picture we need both these properties, but because of the principle of indeterminacy it is impossible to design an experiment which will show both of them in all detail at the same time. Any experiment will reveal the details of either the wave or the corpuscular character, according to the purpose for which the experiment is designed.